Graphs of Linear Equations - Mark scheme

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

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Question 1

Practice question written for this worksheet (not from a past paper)

The point $(p, 50)$ lies on the line $y = 4x + 2$. Work out p.

$50 = 4p + 2$

$4p = 48$

Answer: p = 12

Question 2

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 6

6				M1	for $y = 3x + c$ oe **or** $y = mx - 2$ oe **or** $3x - 2$ **or** eg $L = 3x - 2$ **or** $y = 3(x \pm a)$
		$y = 3x - 2$	2	A1	oe eg $y - 4 = 3(x - 2)$ $y - 1 = 3(x - 1)$ $y - a = 3(x - b)$ where (a, b) is any coordinate on the line
					Total 2 marks

Question 3

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 9

Question	Working	Answer	Mark	Notes
9		$y = 1.5x - 3$	2	B2 oe accept $y = 1.5x + -3$ oe B1 for $y = 1.5x + c$ (where c can be zero) or $y = mx - 3$ (where m is any value but $m \neq 0$ or 1.5) or $1.5x - 3$ or Gradient = 1.5 oe eg $m = \frac{3}{2}$ or a clear calculation for gradient oe (must be labelled or the meaning shown by their diagram or working)
				Total 2 marks

Question 4

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 10

10		eg $y = -5x (+ k)$ **or** $y - a = -5(x - b)$ **or** eg $y = mx + 6$ **or** $y - 6 = m(x - 0)$ **or** eg $-5x + 6$ **or** $L = -5x + 6$		2	M1 for the equation of any line with gradient -5 other than $y = 2 - 5x$ **or** for the equation of any line passing through the point (0, 6) **or** the correct line missing '$y =$' or with the wrong subject
		Correct answer scores full marks (unless from obvious incorrect working)	$y = -5x + 6$		A1 oe equation eg $y = 6 - 5x$ **or** $y - 6 = -5(x - 0)$ **or** $y + 5x = 6$
					Total 2 marks

Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 12

12	$y = \frac{7 - 5x}{2}$ or $y = \frac{7}{2} - \frac{5}{2}x$ or $y = 3.5 - 2.5x$ or $2y = 7 - 5x$ oe		2	M1 for making y or $2y$ the subject Allow $y = -\frac{5}{2}x + c$ oe
		-2.5		A1 for $-\frac{5}{2}$ or -2.5
				Total 3 marks

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2023 · Question 11

Q	Working	Answer	Mark	Notes
	x −2 −1 0 1 2 3 y 5 3 1 −1 −3 −5 (−2, 5) (−1, 3) (0, 1) (1, −1) (2, −3) (3, −5)	Correct line between $x = -2$ and $x = 3$	3	B3 for a correct line between $x = -2$ and $x = 3$ (B2 for a correct straight line segment through at least 3 of (−2, 5) (−1, 3) (0, 1) (1, −1) (2, −3) (3, −5) **or** for all of (−2, 5) (−1, 3) (0, 1) (1, −1) (2, −3) (3, −5) plotted but not joined) (B1 for at least 2 correct points stated (may be in a table) **or** plotted **or** for a line drawn with a negative gradient through (0, 1) **or** for a line with a gradient of −2)
				Total 3 marks

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2023 · Question 12

	(−1, 6) (0, 4) (1, 2) (2, 0) (3, −2) (4, −4)	For a correct line between $x = -1$ and $x = 4$	3	B3 For a correct line between $x = -1$ and $x = 4$ B2 For a correct straight line segment through at least 3 of (−1, 6) (0, 4) (1, 2) (2, 0) (3, −2) (4, −4) **OR** for all of (−1, 6) (0, 4) (1, 2) (2, 0) (3, −2) (4, −4) plotted but not joined **OR** for a line drawn with a negative gradient B1 through (0, 4) and clear intention to use a gradient of −2 For at least 2 correct points stated (may be in a table) **OR** for a line drawn with a negative gradient through (0, 4) **OR** for a line with a gradient of −2
	Correct answer scores full marks (unless from obvious incorrect working)			**Total 3 marks**

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Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2022 · Question 13

Q	Working	Answer	Mark	Notes
	x −1 0 1 2 3 4 5 y 5 3 1 −1 −3 −5 −7 (−1, 5) (0, 3) (1, 1) (2, −1) (3, −3) (4, −5) (5, −7)	Correct line between $x = -1$ and $x = 5$	3	B3 for a correct line between $x = -1$ and $x = 5$ (B2 for a correct straight line segment through at least 3 of (−1, 5) (0, 3) (1, 1) (2, −1) (3, −3) (4, −5) (5, −7) **or** for all of (−1, 5) (0, 3) (1, 1) (2, −1) (3, −3) (4, −5) (5, −7) plotted but not joined) (B1 for at least 2 correct points stated (may be in a table) **or** plotted **or** for a line drawn with a negative gradient through (0, 3) **or** for a line with a gradient of −2)
				Total 3 marks

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2024 · Question 13

Q	Working	Answer	Mark	Notes
	x −2 −1 0 1 2 3 4 y −7 −5 −3 −1 1 3 5 (−2, −7) (−1, −5) (0, −3) (1, −1) (2, 1) (3, 3) (4, 5)	Correct line between $x = -2$ and $x = 4$	3	B3 for a correct line between $x = -2$ and $x = 4$ (B2 for a correct straight line segment through at least 3 of (−2, −7) (−1, −5) (0, −3) (1, −1) (2, 1) (3, 3) (4, 5) **or** for all of (−2, −7) (−1, −5) (0, −3) (1, −1) (2, 1) (3, 3) (4, 5) plotted but not joined) (B1 for at least 2 correct points stated (may be in a table) **or** plotted **or** for a line drawn with a positive gradient through (0, −3) **or** for a line with a gradient of 2)
				Total 3 marks

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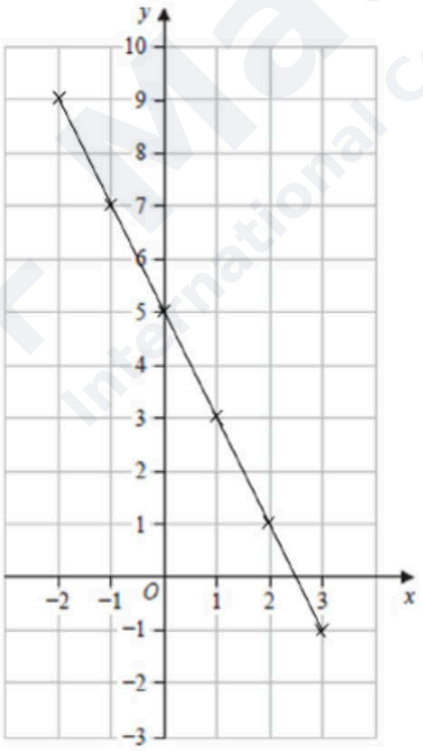
Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2020 · Question 16

Q	Working	Answer	Mark	Notes
	x −2 −1 0 1 2 3 y 15 11 7 3 −1 −5 (−2, 15) (−1, 11) (0, 7) (1, 3) (2, −1) (3, −5)	Correct line between $x = -2$ and $x = 3$	3	B3 for a correct line between $x = -2$ and $x = 3$ (B2 for a correct straight line segment through at least 3 of (−2, 15) (−1, 11) (0, 7) (1, 3) (2, −1) (3, −5) **or** for all of (−2, 15) (−1, 11) (0, 7) (1, 3) (2, −1) (3, −5) plotted but not joined) (B1 for at least 2 correct points stated (may be in a table) or plotted or for a line drawn with a negative gradient through (0, 7) or for a line with a gradient of −4)
				Total 3 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 16

Question	Working	Answer	Mark	Notes
	x −2 −1 0 1 2 3 y 9 7 5 3 1 −1 (−2, 9) (−1, 7) (0, 5) (1, 3) (2, 1) (3, −1)  	Correct line between $x = -2$ and $x = 3$	3	B3 for a correct line between $x = -2$ and $x = 3$ (B2 for a correct straight line segment through at least 3 of (−2, 9) (−1, 7) (0, 5) (1, 3) (2, 1)(3, −1) **or** for all of (−2, 9) (−1, 7) (0, 5) (1, 3) (2, 1)(3,−1)) plotted but not joined) (B1 for at least 2 correct points stated (may be in a table) or plotted **or** for a line drawn with a negative gradient through (0, 5) **or** for a line with a gradient of −2)
				Total 3 marks

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Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 5

5	x	−2	−1	0	1	2	3	4	Correct line	3	B3 for a correct line between $x = -2$ and $x = 4$ If not B3 then award B2 for a line segment through at least 3 of (−2, 10), (−1, 7.5), (0, 5), (1, 2.5), (2, 0), (3, −2.5), (4, −5) **or** all points plotted correctly If not B2 then award B1 for at least 2 correct points plotted or stated (may be seen in a table) **or** for a line drawn with a negative gradient through (0, 5) **or** for a line with a gradient of −2.5
	y	10	7.5	5	2.5	0	−2.5	−5			
											Total 3 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 9

Q	Working	Answer	Mark	Notes
		$y = -\frac{1}{2}x + 1$	3	B3 for $y = -\frac{1}{2}x + 1$ oe but must be in form $y = mx + c$ eg $y = -0.5x + 1$ (B2 for $(L =) -\frac{1}{2}x + 1$ **or** for $y = -\frac{1}{2}x + c$ **or** for $y = mx + 1$ where $m \neq 0$ **or** for a correct equation in the incorrect form eg $2y + x = 2$) (B1 for $(L =) -\frac{1}{2}x + c$ **or** $(L =) mx + 1$ where $m \neq 0$ **or** $m = -\frac{1}{2}$ **or** gradient $= -\frac{1}{2}$ oe eg $\frac{3-(-1)}{-4-4}$)
				Total 3 marks

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Question 14 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2022 · Question 10*

				sign between them or −2 as an answer
				Total 2 marks

10	$3\div 2$ (=1.5 or $\frac{3}{2}$) **or** eg $\frac{5--1}{4(-0)}$ **or** $c=-1$		3	M1	for correct method to find gradient or the correct value of c for gradient, may see a correct calculation **or** $\frac{3}{2}$ oe **or** $1.5x\ (+c)$ oe for value of c, allow $c=-1$, $y=-1$, $(L=)\ mx-1$ oe
	$y=$ "1.5"$x\ (+c)$ **or** $y=mx-1$ **or** eg $y-5=m(x-4)$			M1	for use of $y=mx+c$ with either m or c correct (NB: $m\neq 0$) **or** for $(L=)\ 1.5x-1$ oe
		$y=\frac{3}{2}x-1$		A1	oe eg $y=1.5x-1$
					Total 3 marks

Question 15 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2019 · Question 10*

10	$3\div 2$ (=1.5) or eg $\frac{4-1}{2(-0)}$ or $c=1$ $y=$ "1.5"$x+c$ **or** $y=mx+1$ or eg $y-4=m(x-2)$	$y=1.5x+1$ oe	3	M1 M1 A1	for correct method to find gradient – may see this on grid. For $c=1$, could be $(L=)\ mx+1$ oe or for $1.5x+c$ for use of $y=mx+c$ with either m or c or for $(L=)\ 1.5x+1$ oe eg $y-4=\frac{3}{2}(x-2)$

Question 16 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2019 · Question 11*

11		Gradient = (−)4 ÷ 2 oe	$y=-2x-1$ oe	3	M1 A2	Correct method to work out the gradient (±) accept $4\div 2$ oe or "m" = 2 If not A2 then A1 for $L=-2x-1$ or $-2x-1$ or $y=2x-1$ or $y=-2x+c$
						Total 3 marks

Question 17 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2016 · Question 12*

12	$m=-2$	$y=-2x+13$	3	M1	For recognising gradient = −2, eg $y=-2x+c$ where $c\neq 4$	M2 for $7=-2\times 3+c$
	$c=13$			M1indep	$c=13$	
				A1	oe e.g. $y-7=-2(x-3)$ SCB2 for $-2x+13$	
						Total 3 marks

Question 18 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 25*

Q	Working	Answer	Mark	Notes
(a)	eg $2y=-7x(+10)$		2	M1 for $2y=-7x(+10)$ or an answer of $-3.5x$ oe or an answer of 3.5 oe
		−3.5		A1 oe
(b)		(0, 5)	1	B1 cao
				Total 3 marks

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2021 · Question 13

Q	Working	Answer	Mark	Notes
(a)	$\frac{8--4}{5-9}\left(=\frac{12}{-4}\right)$ oe or $\frac{-4-8}{9-5}\left(=\frac{-12}{4}\right)$ oe		2	M1 condone correct gradient embedded in an equation e.g. $y=-3x+c$ or expression e.g. $-3x$ or for an answer of 3
		-3		A1
(b)		$\frac{1}{4}$	1	B1 accept 0.25 or $-\frac{1}{-4}$ oe
				Total 3 marks

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2023 · Question 13

13	eg (gradient =) $\frac{12--48}{-5-19}\left(=-2.5\right)$ oe		3	M1 for a method to find the gradient
	eg $12=$ "-2.5" $\times -5+c$ oe $y-12=$ "-2.5"$(x--5)$ oe			M1 ft their gradient
	Correct answer scores full marks (unless from obvious incorrect working)	$y=-2.5x-0.5$		A1 oe eg $y-12=-2.5(x+5)$ **or** $2y+5x+1=0$
				Total 3 marks

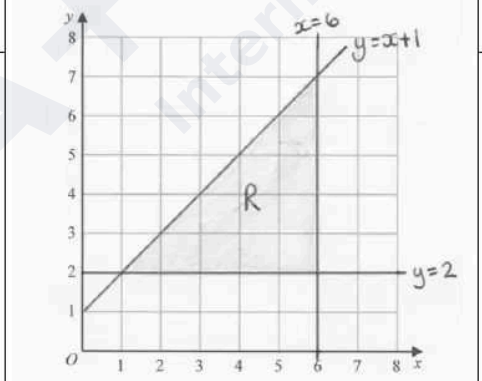
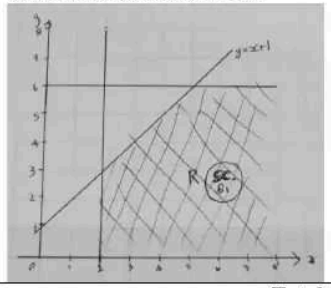
Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 20

20	Gradient of **M** $=-\frac{1}{4}$ oe		3	M1 for a statement that the gradient of **M** is $-\frac{1}{4}$, may be implied by equation of line with gradient of $-\frac{1}{4}$
	eg $1=$ "$-\frac{1}{4}$"$\times 8+c$ or $y-1=$ "$-\frac{1}{4}$"$(x-8)$			M1 ft a gradient of $\frac{1}{4}$ or -4 for this mark, substitution of the given coordinate to find the equation
	Correct answer scores full marks (unless from obvious incorrect working)	$y=-\frac{1}{4}x+3$		A1 oe $y=-0.25x+3$ must be in the form $y=$
				Total 3 marks

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 2

2 (a)(i)		$y=2$ drawn	3	B1	Lines (can be solid, dotted or dashed) must be at least 2 cm long and need not be labelled
(ii)		$x=6$ drawn		B1	
(iii)		$y=x+1$ drawn		B1	
(b)			1	B1ft	Correct region indicated ft dep on at least B2 scored in (a) and a vertical line, a horizontal line and a diagonal line with a positive gradient SCB1 for $y=x+1$, $y=6$ and $x=2$ and area shaded as shown below 
					Total 4 marks

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2015 · Question 10

10. (a)	$y=\frac{2x+6}{3}$ or $y=\frac{2}{3}x+2$ e.g. $\frac{4-2}{3-0}$			M1	For a correct equation with y the subject or $\frac{\text{diff of } y \text{ values}}{\text{diff of } x \text{ values}}$ for any two correct points on the line. Allow 0.666(66666....) rounded or truncated to at least 3SF instead of $\frac{2}{3}$
		$\frac{2}{3}$	2	A1	Allow M1A1 if $\frac{2}{3}$ follows an equation with an incorrect constant. Eg $y=\frac{2}{3}x+3$ followed by an answer of $\frac{2}{3}$ M1A0 for $\frac{2}{3}x+2$ or $\frac{2}{3}x$ Allow 0.666(66666....) rounded or truncated to at least 3SF instead of $\frac{2}{3}$
(b)	$5=-3\times2+c$ or $c=5+3\times2$			M1	
		$y=-3x+11$	2	A1	Allow ISW oe [e.g. $y-5=-3(x-2)$]
					Total 4 marks

Question 24

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2016 · Question 13

13	e.g. $\frac{9-3}{6--2}\left(=\frac{3}{4}\right)$		5	M1	for method to find gradient of **L**	
	$y=$ "0.75"$x+c$			M1	use of their gradient in an equation c may be numerical	M2 for $y--1=$ "$\frac{3}{4}$"$(x-5)$ oe
	$-1=$ "0.75"$\times 5+c$ $\left(c=-\frac{19}{4}\right)$			M1	method to find c	
	$y=\frac{3}{4}x+-\frac{19}{4}$ oe			A1	correct equation (equation in any form)	
		$4y-3x=-19$		A1	oe with integer coefficients e.g. $3x-4y=19$; $4y=3x-19$	
					Total 5 marks	

CHALLENGE QUESTIONS

Question 25 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2017 · Question 12*

12	(a)	$2y=3x-15$ or $-2y=15-3x$ or $1.5x-y=7.5$		3	M1	Or for finding the coordinates of two correct points that lie on the line
		$y=1.5x-7.5$ or $y=\frac{3x-15}{2}$ or $y=\frac{15-3x}{-2}$ oe			M1	or $\frac{\text{difference of } y \text{ values}}{\text{difference of } x \text{ values}}$ for any two correct points on the line
			1.5		A1	oe Do not penalise a mistake in the constant term if the correct answer is given. SCB2 for $1.5x$ SCB1 ft from their $y=ax+b$
	(b)		(0, −7.5)	1	B1	oe
	(c)	$0=1.5\times-2+c$ or $3\times-2=k$ or $y-0=1.5(x--2)$		2	M1	ft 1.5 from (a) or $c=3$
			$y=1.5x+3$		A1	ft 1.5 from (a) or $3x-2y=-6$ or $y=1.5(x+2)$ oe
						Total 6 marks

Question 26 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 25*

Q	Working	Answer	Mark	Notes
	$\left(\frac{4+8}{2},\frac{7-5}{2}\right)$ oe or (6, 1)		5	M1 for finding the midpoint of PR
	$\frac{7--5}{4-8}\left(=-\frac{12}{4}=-3\right)$ oe			M1 for method to find the gradient of PR
	"−3"$\times m=-1$ oe or $(m=)\frac{-1}{\text{"}-3\text{"}}$ or $(m=)\frac{1}{3}$			M1ft for finding the gradient of QS, may be seen embedded in an equation, ft their gradient of PR
	"1" = "$\frac{1}{3}$"("6") + c oe or $c=-1$ **or** $y-1=\frac{1}{3}(x-6)$ **or** $y=\frac{1}{3}x-1$			M1ft (dep on previous M1) for finding the equation through QS, ft their gradient of PR and their midpoint of PR, do not allow (4, 7) or (8, −5) as midpoint PR
	Correct answer scores full marks (unless from obvious incorrect working)	$3y=x-3$		A1 oe eg $6y=2x-6$ or $3y-x+3=0$ etc but must be integer coefficients accept $a=3$, $b=1$, $c=-3$
				Total 5 marks

Question 27 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2020 · Question 22*

Question	Working	Answer	Mark	Notes
	gradient of $JK=-0.5$ **or** $m\times 2=-1$		6	M1 for finding the gradient of JK using $m_1\times m_2=-1$
	$\frac{k-15}{6-j}=-\frac{1}{2}$ **or** $2k-j=24$ **or** $j=2k-24$ **or** $k=\frac{j+24}{2}$ oe			M1 for expressing the gradient of JK in terms of j and k or a correct equivalent equation
	$(j-6)^2+(k-15)^2=80$ oe			M1 for finding equation

	or $\left(\frac{j+6}{2}, \frac{k+15}{2}\right)$ oe **or** $(j+4)^2+196=100+(k-1)^2$ oe			of JK in terms of j and k **or** for finding the midpoint of M **or** for equating length HJ with length HK
	eg $3k^2-78k+495=0$ oe **or** $5j^2-60j-140=0$ oe **or** $5k^2-150k+1045=0$ oe **or** $3j^2-12j-36=0$ oe **or** gradient HM: eg $\frac{\frac{k+15}{2}-1}{\frac{j+6}{2}+4}=2$ **or** $k=2j+15$ **or** $j=\frac{k-15}{2}$ oe			M1 (dep on M3) writing a correct quadratic expression in the form ax^2+bx+c $(=0)$ (allow $ax^2+bx=c$) **or** A correct equation for the gradient of HM in terms of j and k or a correct equivalent equation
	eg $(k-15)(k-11)(=0)$ **or** $\frac{78\pm\sqrt{(-78)^2-4\times3\times495}}{2\times3}$ **or** $(k-13)^2-169+165(=0)$	eg $(j-6)(j+2)(=0)$ **or** $\frac{12\pm\sqrt{(-12)^2-4\times3\times-36}}{2\times3}$ **or** $(j-2)^2-4-12(=0)$		M1 (dep on M3) for a complete method to solve their 3-term quadratic equation (allow one sign error in the use of the quadratic formula) **or** a correct method to eliminate either j or k eg $2k-24=\frac{k-15}{2}$ oe **or** $\frac{j+24}{2}=2j+15$ oe
	$j=-2, k=11$			A1
				Total 6 marks

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Question		Working	Answer	Mark	Notes
		$\left(\frac{j+6}{2}, \frac{k+15}{2}\right)$ oe		6	M1 for finding the midpoint of M
ALT		$\frac{\frac{k+15}{2}-1}{\frac{j+6}{2}+4} = 2$ **or** $k-2j=15$ **or** $k=2j+15$ **or** $j=\frac{k-15}{2}$ oe			M1 for expressing the gradient of JK in terms of j and k or a correct equivalent equation
		$(j-6)^2+(k-15)^2=80$ oe **or** $(j+4)^2+196=100+(k-1)^2$ oe			M1 for finding the length of JK in terms of j and k **or** for equating length HJ with length HK
		E.g. $5j^2-12j-44=0$ **or** $3j^2+48j+84=0$ oe; E.g. $5k^2-174k+1309=0$ **or** $3k^2+6k-429=0$ oe			M1 (dep on M3) writing the correct quadratic expression in form $ax^2+bx+c\ (=0)$ allow $ax^2+bx=c$
		E.g. $(5j-22)(j+2)(=0)$ **or** $\frac{12\pm\sqrt{(-12)^2-4\times5\times-44}}{2\times5}$ **or** $(j+8)^2-64+28(=0)$; E.g. $(5k-119)(k-11)(=0)$ **or** $\frac{174\pm\sqrt{(-174)^2-4\times5\times1309}}{2\times5}$ **or** $(k+1)^2-1-143(=0)$			M1 (dep on M3) for a complete method to solve their 3-term quadratic equation (allow one sign error in the use of the quadratic formula)
		$j=-2,\ k=11$			A1
					Total 6 marks

Question 28 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 23*

Q	Working	Answer	Mark	Notes
	$y=\frac{3}{5}x\left(+\frac{6}{5}\right)$ **or** $y=0.6x(+1.2)$ **or** (gradient =) $\frac{3}{5}$ or 0.6 oe		6	M1 for correct gradient which may be seen in an equation. Condone $\frac{3}{5}x$ or $0.6x$
	$k=-2$			B1 for $k=-2$
	$\left(\frac{"-2"+8}{2}, \frac{1+7}{2}\right)$ oe or (3, 4)			M1 for finding the midpoint (use of their k where $k<0$)
	$"\frac{3}{5}"m=-1$ **or** $(m=)"-\frac{5}{3}"$			M1ft their gradient for use of $m_1\times m_2=-1$ Allow $-\frac{5}{3}=-1.67$ or better
	$"4"="-\frac{5}{3}"\times"3"+c$ or $c=9$ or $y-"4"="-\frac{5}{3}"(x-"3")$			M1 dep on M3
	Working required	$5x+3y=27$		A1 allow equation in any form where p, q and r are integers
				Total 6 marks

Question 29

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23	$\left(\frac{-1+5}{2}, \frac{6-4}{2}\right)$ or $\left(\frac{4}{2}, \frac{2}{2}\right)$ or (2, 1)		6	M1 for finding midpoint
	$\frac{-4-6}{5--1}$ or $\frac{6--4}{-1-5}$ or $-\frac{10}{6}$ or $-\frac{5}{3}$			M1 indep for finding the gradient of PQ
	$\frac{-1}{-\frac{10}{6}}$ or $\frac{6}{10}$ or $\frac{-1}{-\frac{5}{3}}$ or $\frac{3}{5}$ or 0.6			M1 for finding the perpendicular gradient to PQ (ft their stated gradient)
	$1=\frac{3}{5}(2)+c$ or $c=-\frac{1}{5}$ or $c=-\frac{2}{10}$ or $c=-0.2$			M1 dep on 1st and 3rd M1 for substituting '(2, 1)' into $y='\frac{3}{5}'x+c$ or find the value of c oe eg $y-'1'='\frac{3}{5}'(x-'2')$
	$y=\frac{3}{5}x-\frac{1}{5}$ or $y=0.6x-0.2$ or $5y=3x-1$			A1 for a correct equation in any form
		$3x-5y-1=0$		A1 for $3x-5y-1=0$ or $5y-3x+1=0$ or $6x-10y-2=0$ oe accept in the form $ax+by=-c$ eg $3x-5y=1$ or $5y-3x=-1$ oe
				Total 6 marks

Alternative Mark Scheme for Q23

23	$(x+1)^2+(y-6)^2$ or $(x-5)^2+(y+4)^2$		6	M1
	$(x+1)^2+(y-6)^2=(x-5)^2+(y+4)^2$			M1 using $PA^2=QA^2$ (for some point A on the line)
	$x^2+2x+1+y^2-12y+36$ or $x^2-10x+25+y^2+8y+16$			M1
	$x^2+2x+1+y^2-12y+36=x^2-10x+25+y^2+8y+16$			M1
	eg $2x+1-12y+36=-10x+25+8y+16$ or $12x+37=20y+41$			A1 for a correct linear equation in x and y
		$3x-5y-1=0$		A1 for $3x-5y-1=0$ oe
				Total 6 marks

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